Akshai Narayan Srinivasan

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Education

University of Washington Seattle, WA
Dual Degree: BS in Computer Engineering, BS in Applied Mathematics *June 2027*

Relevant Coursework: Machine Learning, Systems Programming, Digital Design, Data Structures & Parallelism, Hardware Software Interface, Discrete Mathematics & Probability, Dynamical Systems, Applied Linear Algebra

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, Dart
Cloud & Databases: AWS (Lambda, API Gateway, S3, DynamoDB, Bedrock), PostgreSQL, MongoDB, MySQL
Frameworks & Tools: ROS2, PyTorch, Isaac Lab, FastAPI, Claude API, Flask, WebSockets, Next.js, Stanford CoreNLP, Git

Experience

General Motors: Cruise Sunnyvale, CA
Software Engineering Intern *June 2026 - Sep 2026*

- Developing **autonomous vehicle** infrastructure for a web-based platform visualizing multi-source **ROS data streams**
- Engineered a **diagnostic telemetry tool** that generates failure reports, provisions **GCS buckets via Terraform** for log retention, and dispatches **Slack** alerts for developer triage
- Shipped **4 features** streamlining topic tree navigation for **800+ users**, including availability filtering for node inputs based on **ROS bag** contents

Washington Embodied Intelligence and Robotics Development Lab Seattle, WA
Undergraduate Research Assistant *Feb 2026 – June 2026*

- Trained **sim2real reinforcement learning** policies for **zero-shot transfer** of contact-rich robotic assembly tasks
- Leveraged **PPO** and an **asymmetric actor-critic** setup in **Isaac Sim** to train **MLP policy networks** on history-conditioned observation spaces
- Scaled **domain randomization** across aggressive physical dynamics, leveraging **curriculum learning**, **ADR**, and **LSTM architectures** to stabilize policy convergence

Neptune Medical Inc. Burlingame, CA
Robotic Software Engineering Intern *Jun 2025 – Sep 2025*

- Developed controls software for **robotic endoscope** using **ROS** and **Python**, orchestrating communication between controllers, sensors, and actuators
- Built **C++ fault detection framework** for haptic input device with automatic failsafe protocols, preventing unsafe robot operations during **live procedures**
- Designed **tendon saturation algorithm** with integrated **haptic feedback** that dynamically generates resistive forces when tendons reach operational limits, preventing mechanical damage to endoscope

DubHacks Seattle, WA
Director of Technology *Jan 2025 – Present*

- Overseeing all software development and infrastructure for the **largest collegiate hackathon** in the Pacific Northwest **(1,200+ participants)** using **FastAPI, Next.js, TypeScript, and React**
- Engineered high-performance judging platform with **OAuth2/JWT authorization**, leveraging **WebSockets** and event-driven architecture to achieve **sub-100ms** latency for 150+ concurrent sessions
- Architected scalable **MongoDB** cluster handling **376K edge requests** and **122K serverless function** invocations during peak hackathon traffic with **100% uptime** and zero data loss

Sample Projects (Portfolio: akshaisrin.live)

DuckRL | *PyTorch, MuJoCo, Reinforcement Learning, PPO* | https://akshaisrin.live/DuckRLWebsite June 2026

- Engineered custom **reinforcement learning** locomotion policies in **PyTorch** and **MuJoCo sim** for **bipedal droids**
- Trained **PPO** actor-critic using **MLP policy networks** with clipped objective functions to output joint positions
- Structured a **two-stage curriculum** for **domain randomization**, leveraging **Rapid Motor Adaptation** to condition the policy on privileged dynamics as physical difficulty scaled
- **Outperformed baseline policies** by maintaining dynamic gait stability across 3 stress tests evaluating payload offsets, push recovery, and friction variations